

```
#empty_list=[  
my_list=[]  
  
#list of integers  
my_list=[1,2,3]  
  
#list with mixed data types  
my_list=[1,"Hello",3.4]  
  
#multiple data type assign  
n_list=[2.9,3000,'Hello',"Welcome to python list"true']  
print(n_list)
```

```
#Nested List Example1  
n_list=["Happy",[2,0,1,5]]  
  
#Nested indexing  
print(n_list[0][3])  
print(n_list[1][2])
```

```
#Nested List Example2  
my_list=["mouse",[8,4,6],['a','b','c']]  
print(my_list)  
print(my_list[0])  
  
#Nested indexing  
print(my_list[1][1])  
print(my_list[2][2])
```

```
#Negative indexing lists
```

```
my_list=['p','r','o','b','e']
```

```
print(my_list[-1])
```

```
print(my_list[-5])
```

```
#Appending and Extending lists in python
```

```
add=[1,3,5]
```

```
#List Lenth & Remove
```

```
thislist=["apple","banana","cherry"]
```

```
thislist.remove("banana")
```

```
print(thislist)
```

```
add.append(7)
```

```
print(add)
```

```
add.extend([9,11,13])
```

```
print(add)
```

```
#concatenating and repeating lists
```

```
add=[1,3,5]
```

```
print(add+[9,7,5])
```

```
print(["re"]*3)
```

```
#Remove Specified Item
```

```
thislist=["apple","banana","cherry"]
```

```
thislist.remove("banana")
```

```
print(thislist)

#Python-Loop List

thislist=["apple","banana","cherry"]

for x in thislist:

    print(x)
```

```
#Sort list

thislist=[100,50,65,82,23]

thislist.sort()

print(thislist)
```

```
#Copy List

thislist=["apple","banana","cherry"]

print(my_list)
```

```
#insert method

fruits=['apple','banana','cherry']

fruits.insert(1,"orange")

print(fruits)
```

```
#Clear the List

thislist=["apple","banana","cherry"]

thislist.clear

print(thislist)
```

#List Count() Method

```
fruits=["apple","banana","cherry","apple"]
```

```
x=fruits.count("apple")
```

```
print(x)
```

#list copy() method

```
list=[10,20,30,40]
```

#Create a copy of the list

```
new_list = list.copy()
```

```
print("Original List:",list)
```

```
print("Copied List:",new_list)
```

#list index() method

```
list=[10,20,30,40,50,]
```

```
value=30
```

```
position=list.index(value)
```

```
print("List:"list')
```

```
print("value:",value)
```

```
print("Index:",position)
```

OUTPUT

```
[2.9, 3000, 'Hello', 'Welcome to python listtrue']
```

```
p
```

```
1
```

```
['mouse', [8, 4, 6], ['a', 'b', 'c']]
```

mouse

4

c

e

p

['apple', 'cherry']

[1, 3, 5, 7]

[1, 3, 5, 7, 9, 11, 13]

[1, 3, 5, 9, 7, 5]

['re', 're', 're']

['apple', 'cherry']

apple

banana

cherry

[23, 50, 65, 82, 100]

['p', 'r', 'o', 'b', 'e']

['apple', 'orange', 'banana', 'cherry']

['apple', 'banana', 'cherry']

2

Original List: [10, 20, 30, 40]

Copied List: [10, 20, 30, 40]

List:list

value: 30

Index: 2